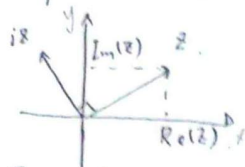


Chap 2.

Def. $\mathbb{C} \cong \mathbb{R}^2$. $z = x + iy \in \mathbb{C} \rightarrow (x, y) \in \mathbb{R}^2$.



$\Rightarrow z \rightarrow \bar{z}$ 相当于在复平面 逆时针 $\frac{\pi}{2}$.

$$\text{i.e. } R_{\frac{\pi}{2}} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -y \\ x \end{pmatrix}.$$

$$\bar{z} = x - iy.$$

$$|z| = \sqrt{z\bar{z}} = \sqrt{x^2 + y^2}.$$

$$d(z, w) = |z - w|.$$

$$\Rightarrow \begin{cases} \text{极. } z = x + iy. \\ \text{极. } z = r(\cos \theta + i \sin \theta). \\ \text{Euler. } z = r \cdot e^{i\theta}. \end{cases}$$

$$R_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

$$z = a + ib$$

$\Leftrightarrow$ 极坐标

$$z = r(\cos \theta + i \sin \theta)$$

$$r = |z|.$$

辐角 $\theta = \arg z$.

$$\cos \theta = \frac{a}{r}.$$

$$\sin \theta = \frac{b}{r}.$$

Theo. 2.1.

$$\textcircled{1}. \bar{\bar{z}} = z. \quad \bar{z} = r \cdot e^{-i\theta}$$

$$z^{-1} = \frac{\bar{z}}{z^2}.$$

$$\textcircled{2}. |Re(z)| \leq |z|, |Im(z)| \leq |z|, |\bar{z}| = |z|.$$

$$|zw| = |z||w|.$$

$$|z+w| \leq |z| + |w|$$

$$|z-w| \geq ||z| - |w||.$$

Def. $c \in \mathbb{C}, r > 0 \in \mathbb{R}$.

开邻域. Open neighborhood $N(c, r) := \{z \in \mathbb{C}, |z - c| < r\}$.

闭邻域. Closed disc $\bar{N}(c, r) := \{z \in \mathbb{C}, |z - c| \leq r\}$.

圆. circle $K(c, r) := \{z \in \mathbb{C}, |z - c| = r\} = \partial N(c, r) = \partial \bar{N}(c, r)$.

穿孔圆盘. Punctured disc $P'(c, r) := \{z \in \mathbb{C}, 0 < |z - c| < r\} = N(c, r) \setminus \{c\}$.

Def.

[Open subset] $\forall u \in U, \exists \delta > 0$ s.t. $N(u, \delta) \subset U$.

[Closed subset]. $C \setminus U = U^c \cap C$.

Def. [complex function]. $f: \mathbb{C} \rightarrow \mathbb{C}, z = x + iy$ 有 $f(z) = u(x, y) + iv(x, y)$,

$$\begin{cases} u = Re(f): \mathbb{R}^2 \rightarrow \mathbb{R}. \\ v = Im(f): \mathbb{R}^2 \rightarrow \mathbb{R}. \end{cases}$$

$$|f|: \mathbb{C} \rightarrow \mathbb{R}. |f|(z) := |f(z)| = \sqrt{u(x, y)^2 + v(x, y)^2}$$

Def. [极限]. $f: \mathbb{C} \rightarrow \mathbb{C}. \forall \varepsilon > 0, \exists \delta > 0$ s.t. $\forall z \in D'(c, \delta),$

$$|f(z) - l| < \varepsilon \Rightarrow \lim_{z \rightarrow c} f(z) = l.$$

Def. [连续]. $f: \mathbb{C} \rightarrow \mathbb{C}. \forall c \in U, \lim_{z \rightarrow c} f(z) = f(c) \Rightarrow f$ 在 U 连续.

Def. [无穷远极限]. $f: \mathbb{C} \rightarrow \mathbb{C}. \forall \varepsilon > 0, \exists K > 0$ s.t. $\forall |z| > K, |f(z) - L| < \varepsilon. \Rightarrow \lim_{z \rightarrow \infty} f(z) = L.$

Def. [无穷极限]. $\forall K \in \mathbb{R}, \exists D > 0$ s.t. $\forall |z| > D, |f(z)| > K \Rightarrow \lim_{z \rightarrow \infty} f(z) = \infty$

Chap 3.

从复平面各方向趋相等.

[可微]. Def. $f: C \rightarrow C$. $f'(c) = \lim_{z \rightarrow c} \frac{f(z) - f(c)}{z - c}$ 存在 $\Rightarrow f$ 在 c 可微.

$$f'(c) := \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}$$

Theo 3.1 线性. $f' = u' + v'$.

$$\text{取 } f' = u'v + uv'. \text{ 商 } f' = \frac{vu' - uv'}{(v)^2}$$

$$\textcircled{3}. f = u(v) \Rightarrow f'(z) = u'(v(z)) \cdot v'(z).$$

$$\textcircled{4}. \text{Poly. } a_0, \dots, a_n \in C \Rightarrow f(z) = a_n z^n + \dots + a_1 z + a_0 \in C.$$

$$\textcircled{5}. \text{Rational. } p(z), q(z) \text{ 无公因子多项式} \Rightarrow f(z) = \frac{p(z)}{q(z)} \in C. (q(z) \neq 0) \quad \left. \begin{array}{l} \text{poly. relation 在 } C \text{ 可微.} \end{array} \right\}$$

Theo 3.2. [Cauchy-Riemann].

$$f(x+iy) = u(x,y) + i v(x,y) \text{ 在 } C \text{ 可微} = \alpha + i\beta \text{ 可微} \Rightarrow \text{在 } (a,b) \text{ 有} \left\{ \begin{array}{l} \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \\ \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} \end{array} \right.$$

Theo 3.3. $f(z) = \underbrace{u(x,y)}_{Re} + i \underbrace{v(x,y)}_{Im}$. $D = N(c, R) \subset D_f \subset C$.

$$\textcircled{1} f \text{ 在 } c \text{ 有 } C-R$$

$$\textcircled{2} u_x, u_y, v_x, v_y \text{ 在 } D \text{ 连续} \quad \left. \begin{array}{l} \end{array} \right\} \Rightarrow f \text{ 在 } c \text{ 可微.}$$

$$\text{可导} \Rightarrow C-R.$$

$$C-R + \text{偏导连续} \Rightarrow \text{可微}$$

$$\text{定义证可微: } \lim_{h \rightarrow 0} \frac{f(z+h) - f(z)}{h} \text{ 存在.}$$

Def. [解析. holomorphic]. f 在 U 可微 $\Rightarrow f$ 在 U 解析.

[全. entire]. f 在 C 可微 $\Rightarrow f$ 是全函数.

证可导: ① 定义 ② $C-R$ + 偏导连续 $\Rightarrow$ 可导

③ 解析条件 ④ 写成 Taylor.

$$\textcircled{5} \frac{\partial f}{\partial \bar{z}} = 0$$

Theo 3.4. U 开. $f: U \rightarrow C$.

$$\textcircled{1} f \text{ 在 } U \text{ 有 } C-R.$$

$$\textcircled{2} u_x, u_y, v_x, v_y \text{ 在 } U \text{ 连续} \quad \left. \begin{array}{l} \end{array} \right\} \Rightarrow f \text{ 在 } U \text{ 解析.}$$

Theo 3.5. $f: U \rightarrow C$. f 在 $c \in U$ 可微 $\Leftrightarrow \frac{\partial f}{\partial \bar{z}} = 0$. 在解析处有 $f'(c) = \frac{\partial f}{\partial z}(c)$.

Theo 3.6. f 在 $U = N(a, R)$ 解析. $\forall z \in U, f'(z) = 0 \Rightarrow \forall z \in U, f'(z) = \text{常数 } c$.

Theo 3.7. f 在 $U = N(a, R)$ 解析, $|f|$ 在 U 常函数 $\Rightarrow f$ 在 U 常函数.

Theo 3.8. [Goursat's Lemma]. $f: U \rightarrow C$.

$$f \text{ 在 } U \text{ 解析} \Rightarrow \exists \psi(z, c) \text{ s.t. } f(z) = f(c) + (z-c)f'(c) + (z-c)\psi(z, c), \\ c \in U, \lim_{z \rightarrow c} \psi(z, c) = 0.$$

Theo 3.9. [Converse Goursat's]. $f: U \rightarrow C$.

$$\exists \gamma \in C \text{ s.t. } \lim_{z \rightarrow c} \frac{f(z) - f(c) - (z-c)\gamma}{z-c} = 0 \Rightarrow f \text{ 在 } c \text{ 可导且 } f'(c) = \gamma.$$

Chap 4.

Def. [收敛和 converge to sum]. $\sum_{n=1}^{\infty} z_n \in \mathbb{C}$ 收敛和为 $S := \lim_{N \rightarrow \infty} S_N = \lim_{N \rightarrow \infty} \sum_{n=1}^N z_n = S$.

[发散] $\sum_{n=1}^{\infty} z_n \in \mathbb{C}$ 不收敛 $\Rightarrow \sum_{n=1}^{\infty} z_n$ 发散.

[绝对收敛 absolutely convergent]. $\sum_{n=1}^{\infty} |z_n| \in \mathbb{C}$ 收敛 $\Rightarrow \sum z_n$ 收敛.

Theo 4.1. [Comparison Test].

[Ratio Test]. $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \begin{cases} < 1, \text{绝对收敛} \\ > 1, \text{发散} \end{cases}$

① $\sum a_n$ 绝对收敛, $|z_n| \leq |a_n| \forall$ 足够大 $n \Rightarrow \sum z_n$ 绝对收敛.

② $\sum a_n$ 发散, $\forall$ 足够大 $n |z_n| \geq |a_n| \Rightarrow \sum z_n$ 发散.

Def. [幂级数 power series]. $\sum_{n=0}^{\infty} C_n (z-a)^n, z, a, C_n \in \mathbb{C}$.

[收敛半径]. $\sum_{n=0}^{\infty} C_n (z-a)^n$ 在 $|z-a| < R$ 收敛,
在 $|z-a| > R$ 发散.

Theo 4.2. $\sum_{n=0}^{\infty} C_n (z-a)^n$ 的 $R \neq 0 \Rightarrow f(z) := \sum_{n=0}^{\infty} C_n (z-a)^n$ 在 $N(a, R)$ 解析,

$$\text{且 } f'(z) = \sum_{n=1}^{\infty} n C_n (z-a)^{n-1}.$$

Def. [Exp.]. $e^z := \sum_{n=0}^{\infty} \frac{z^n}{n!} = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots, z \in \mathbb{C}$.

$$(e^z)' = e^z.$$

$$e^{z+w} = e^z \cdot e^w.$$

Def. [Trig. Hyperbolic] $\rightarrow$ 在 $\mathbb{C}$ 解析.

$$\cos z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!} = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \dots$$

$$\sin z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!} = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \dots$$

$$\cosh z = \sum_{n=0}^{\infty} \frac{z^{2n}}{(2n)!} = 1 + \frac{z^2}{2!} + \frac{z^4}{4!} + \dots$$

$$\sinh z = \sum_{n=0}^{\infty} \frac{z^{2n+1}}{(2n+1)!} = z + \frac{z^3}{3!} + \frac{z^5}{5!} + \dots$$

$$\cos^2 t = \frac{1 + \cos(2t)}{2}$$

$$\sin^2 t = \frac{1 - \cos(2t)}{2}$$

Theo 4.3. $\forall z \in \mathbb{C},$

$$e^{iz} = \cos z + i \sin z, e^z = \cosh z + \sinh z.$$

$$\cos z = \frac{1}{2} (e^{iz} + e^{-iz}), \sin z = \frac{1}{2i} (e^{iz} - e^{-iz}), \cos^2 z + \sin^2 z = 1.$$

$$\cosh z = \frac{1}{2} (e^z + e^{-z}), \sinh z = \frac{1}{2} (e^z - e^{-z}).$$

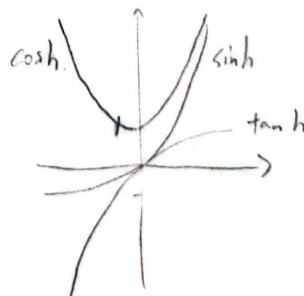
Theo 4.4.

$$(\cos z)' = -\sin z, (\sin z)' = \cos z, (\cosh z)' = \sinh z, (\sinh z)' = \cosh z.$$

Theo 4.5.

$$\cos(z+w) = \cos z \cos w - \sin z \sin w$$

$$\sin(z+w) = \sin z \cos w + \cos z \sin w.$$



Theo 4.6 [Periodicity].

$$e^{z+2\pi i} = e^z.$$

复域中 $\log$ 不再双射.

Def. [Principal log]. $\log z = \ln|z| + i \arg z$, $z \in \mathbb{C}$, $\arg z \in (-\pi, \pi]$.

由于辐角，
log z 的表示
可能多样，
不影响。

Def. [多值, multifunction]. $\log$ 的奇点由于是 branch point (绕圈取值, 不连续)
既非正极点

$$\text{Arg } z: z \rightarrow \{\arg z + 2\pi n, n \in \mathbb{Z}\}.$$

$$\text{Log } z: z \rightarrow \{\log z + i(2\pi n), n \in \mathbb{Z}\}.$$

$$\text{Arg}(zw) = \text{Arg } z + \text{Arg } w. \quad \text{Log}(zw) = \text{Log } z + \text{Log } w.$$

Def. [复幂, complex power]. $c, z \in \mathbb{C}$.

$$c^z = e^{z \text{Log } c}$$

$\rightarrow$ 显然为多值函数, 所以取标准值 $c^z = e^{z \log c}$.

Theo 4.7. $(\log z)' = \frac{1}{z}$, $z \in \mathbb{C}, z \neq 0$.

Def. [复开方, n-th root]. $z = r \cdot e^{i\theta}$, $n \in \mathbb{N}$.

$$\text{多值. } z^{\frac{1}{n}} = \left\{ r^{\frac{1}{n}} \cdot e^{i \frac{(\theta + 2k\pi)}{n}}, k = 0, 1, \dots, n-1 \right\}.$$

$$\text{主值. } z^{\frac{1}{n}} = r^{\frac{1}{n}} \cdot e^{i \frac{\theta}{n}}.$$

Def. [可去奇点, removable singularity]. $f: U \rightarrow \mathbb{C}$, $N(c, r) \cap U$.

$\lim_{z \rightarrow c} f(z)$ 存在但在 c 不连续 $\Rightarrow c$ 是可去奇点 $\rightarrow$ 可改变 c 点值使其可微.

Def. [奇点, singularity]. $\lim_{z \rightarrow c} f(z)$ 不存在 $\Rightarrow c$ 是奇点.

[极点, pole]. c 是奇点, $\exists N \exists n \geq 1$ s.t. $\lim_{z \rightarrow c} (z-c)^n f(z) < \infty$. $\Rightarrow c$ 是极点.

[极点阶数, order] 使其为奇点的最小 n .

Def. [亚纯, meromorphic]. f 在开 $H \subset \mathbb{C}$ 除极点外都解析 $\Rightarrow f$ 在 H 亚纯.

Def. [有理]. $p, q: \mathbb{C} \rightarrow \mathbb{C}$, $r: U \rightarrow \mathbb{C}$.

必须证明是 pole! 本性奇点不行

$$p, q \text{ 多项} \Rightarrow r(z) = \frac{p(z)}{q(z)} \text{ 在 } U = \{z \in \mathbb{C}, q(z) \neq 0\} \text{ 有理. } \lim_{z \rightarrow c} |(z-c)^n f(z)| < \infty$$

Theo 4.8. p, q 多项. $h(z) = \frac{p(z)}{q(z)}$ 有理且 p, q 无公因子 $\Rightarrow r(z)$ 亚纯, 极点为 $z \in \mathbb{C}$ s.t. $q(z) = 0$.

Def. [isolated essential singularity, 孤立本质奇点]. 非可去的奇点, 且不为极点.

[non-iso essential singularity, 非孤立本质奇点].

求奇点类型:

① 先看是否可去奇点, i.e. $\lim_{z \rightarrow c} f$ 存在.

② 再看是否极点,

否则本性奇点.

① [奇点]. $\lim_{z \rightarrow c} f$ DNE

[可去奇]. $\lim_{z \rightarrow c} f$ 存在但在 c 不连续.

[极点]. $\exists n \in \mathbb{N}$ s.t. $\lim_{z \rightarrow c} |(z-c)^n f| < \infty$

[本性奇]. 非以上 n .

②. Laurent. $f: \sum_{n=-\infty}^{\infty} a_n (z-c)^n$.

$\forall$ 有限 $n \in \mathbb{N}$ 有 $a_n \neq 0 \Rightarrow c$ 本性奇, 否则.

使得 $a_n = 0$ 的最小 $n = \text{ord}(f, c)$

$\text{ord}(f, c) \geq 0 \Rightarrow c$ 可去奇

$\text{ord}(f, c) < 0 \Rightarrow c$ 极点
阶数 $n = |\text{ord}(f, c)|$

Chap 5.

Def. [角, angle]. 曲线 C_1, C_2 有 $C_1(0) = C_2(0)$, $C_1'(0) \neq 0$, $C_2'(0) \neq 0$.
 C_1 与 C_2 在 0 夹角: $\arg(C_2'(0) - C_1'(0))$.

Theo 5.1 [保角变换, angle preserving].

$f: U \rightarrow \mathbb{C}$ 解析. 曲线 $C_1(t), C_2(t) \subset U$ 且在 $t=0$ 交叉有 $C = C_1(0) = C_2(0)$, $f'(C), C_1'(0), C_2'(0) \neq 0$
 $\Rightarrow C_1, C_2$ 夹角 α 等于 $f(C_1), f(C_2)$ 夹角. i.e. $\alpha = \beta \pmod{2\pi}$.

Def. [保角, conformal]. f 在 U 解析, $\forall c \in U, f'(c) \neq 0 \Rightarrow f$ 在 U 保角.

Theo 5.2. [Riemann Mapping]. $\forall$ 闭包无交叉分段光滑曲线 C , $\exists$ 双射保角 $f: D(1) \rightarrow N(0,1)$
 且 $f^{-1}: N(0,1) \rightarrow D(1)$ 也保角. 闭包曲线内部

Def. [Möbius/Bilinear 变换].

$$f: \mathbb{C}^* \rightarrow \mathbb{C}^*. f(z) = \frac{az+b}{cz+d}, \quad ad-bc \neq 0, \quad a, b, c, d \in \mathbb{C}.$$

$$\mathbb{C}^* = \mathbb{C} \cup \infty$$

Möbius:

① 在 $\mathbb{C} \setminus \{-\frac{d}{c}\}$ 解析

②. 极点为 $z = -\frac{d}{c}$

③. $f'(z) = \frac{ad-bc}{(cz+d)^2}$ 保角.

Theo 5.3.

(1). Möbius 的逆仍 Möbius. Möbius 的复合仍 Möbius.

(2). (M1). $f(z) = az$. $a = re^{i\theta} \neq 0$. 放缩 + 旋转.

(M2). $f(z) = z+b$ 平移.

(M3). $f(z) = \bar{z}$. 复反演 complex inversion: 半径取倒数, 角度取反

每个 Möbius $f(z) = \frac{az+b}{cz+d}$ 都是以上 3 类组合.

(3). Möbius 把 $\Gamma \times \text{圆}$ 映成 $\Gamma \times \text{圆}$.

$\Gamma \times \text{圆}$: 圆或直线.

(4). 对于 3 组不同点 (z_1, z_2, z_3) 和 (w_1, w_2, w_3) , $\exists$ 唯一 Möbius f s.t. $f(z_i) = w_i, i=1, 2, 3$.

共形: 保角保定向. 解析且 $f' \neq 0 \Rightarrow$ 共形. 共形 $\nrightarrow$ Möbius.

Möbius: ① $\Gamma \times \text{圆} \rightarrow \Gamma \times \text{圆}$ ② 在 D 解析 ③ 双射

Chap b.

Def. [有界, bounded]. $S \subset \mathbb{C}$, $\forall z \in S, \exists K > 0$ s.t. $|z| \leq K \Rightarrow S$ 有界. i.e. $S \subset \bar{N}(0, K)$.

Def. [覆盖, covering]. $S \subset \mathbb{C}$, V_i 开.

$S \subset \bigcup_{i \in I} V_i \Rightarrow \mathcal{C} = \{V_i, i \in I\}$ 是 S 的开覆盖. I 有限时为有限覆盖.

[子覆盖, subcovering]. $J \subset I$,

$S \subset \bigcup_{i \in J} V_i \Rightarrow \mathcal{J} = \{V_i, i \in J\}$ 是 $\mathcal{C}$ 的子覆盖.

Def. [紧性, Compactness]. 任意开覆盖都有有限子覆盖的 S .

Theo b.1 [Heine-Borel]. S 紧 $\Leftrightarrow S$ 有界闭.

Theo b.2. $f: U \rightarrow \mathbb{C}$. 紧集 $S \subset U \Rightarrow f$ 在 S 有界且有上确界.

Coro. $f: U \rightarrow \mathbb{C}$. 紧集 $S \subset U$. $\forall s \in S, f(s) \neq 0 \Rightarrow f$ 在 S 下确界 > 0 .

Def. [参数曲线, parametrised].

$\gamma: [a, b] \rightarrow \mathbb{C}$. $\gamma(t) = \alpha(t) + i\beta(t)$.

$\gamma(t) = e^{it}$ 单位圆

[轨迹, image].

$\gamma^* = \{ \gamma(t), t \in [a, b] \}$.

[方向, orientation].

[闭曲线, close]. $\gamma(a) = \gamma(b)$.

[简单曲线, simple]. $a \leq t < t' \leq b$ 时不能 $\gamma(t) = \gamma(t')$ 除非 $t = a$ 且 $t' = b$. 即不自交.

Def. [可求长, rectifiable]. $\mathcal{D}$ 是 $[a, b]$ 分割. $\{L(\gamma^*, D), D \in \mathcal{D}\}$ 有上界 $\Rightarrow \gamma^*$ 可求长.

[长度, length]. γ^* 可求长 $\Rightarrow \Lambda(\gamma^*) = \sup \{L(\gamma^*, D), D \in \mathcal{D}\}$.

Theo b.3 $\gamma^* = \{ \gamma(t), t \in [a, b] \}$. $\operatorname{Re} \gamma, \operatorname{Im} \gamma: [a, b] \rightarrow \mathbb{R}$ 可微且一阶导连续

$\Rightarrow \gamma^*$ 可求长. $\Lambda(\gamma^*) = \int_a^b |\gamma'(t)| dt$.

Theo b.4. $f, g: \mathbb{R} \rightarrow \mathbb{C}, k \in \mathbb{C}$.

Rectifiable.

$$(1) \int_a^b (f \pm g) dt = \int_a^b f dt \pm \int_a^b g dt$$

$$(2) \int_a^b k \cdot f dt = k \int_a^b f dt.$$

$$(3) \int_a^b f dt + \int_b^c f dt = \int_a^c f dt.$$

Def. [沿曲线积分].

$$\int_{\gamma} f(z) dz = \int_a^b f(\gamma(t)) \gamma'(t) dt.$$

Theo b.5. $\gamma: [0, 2\pi] \rightarrow \mathbb{C}$. $\gamma(t) = e^{it}$. $n \in \mathbb{Z}$ 有

$$\int_{\gamma} z^n dz = \begin{cases} 2\pi i, & n = -1 \\ 0, & \text{otherwise.} \end{cases}$$

除 $\frac{1}{z}$ 外其余

幂函数沿单位圆积分为 0.

Def. [逆向曲线], $\gamma: [a, b] \rightarrow \mathbb{C}$ 的逆向曲线: $\bar{\gamma}: [a, b] \rightarrow \mathbb{C}, \bar{\gamma}(t) = \gamma(a+b-t)$.

Theo 6.6. $\gamma: [a, b] \rightarrow \mathbb{C} \Rightarrow \int_{\bar{\gamma}} f(z) dz = - \int_{\gamma} f(z) dz$. 改变定向和为取反

Def. [沿分段光滑曲线积分].

$\gamma: [a, b] \rightarrow \mathbb{C}, a = c_0 < c_1 < \dots < c_m = b, \gamma_i: [c_{i-1}, c_i] \rightarrow \mathbb{C}$ 光滑且 $\gamma_i(c_i) = \gamma_{i+1}(c_i), i=1, \dots, m-1$.

$$\int_{\gamma} f(z) dz = \sum_{i=1}^m \left(\int_{\gamma_i} f(z) dz \right).$$

Theo 6.7. $f: [a, b] \rightarrow \mathbb{C}$ 连续, $F(x) = \int_a^x f(t) dt, x \in [a, b]$.

$\Rightarrow \forall x \in [a, b], F'(x) = f(x)$.

若 $\theta: [a, b] \rightarrow \mathbb{C}$ 有 $\theta' = f$ 则 $\int_a^b f(t) dt = \theta(b) - \theta(a)$.

Theo 6.8. $\gamma: [a, b] \rightarrow \mathbb{C}$ 分段光滑, $F: U \rightarrow \mathbb{C}, \gamma^* \subset U, F'$ 存在且在 γ^* 连续则

复变函数基本定理

$$\int_{\gamma} F'(z) dz = F(\gamma(b)) - F(\gamma(a)).$$

特别的, γ 闭则 $\int_{\gamma} F'(z) dz = 0$.

Coro. $\gamma: [a, b] \rightarrow \mathbb{C}$ 分段光滑且闭, $\Rightarrow \int_{\gamma} z dz = \int_{\gamma} z^2 dz = 0$.

Def. [围道, Contour]. 分段光滑简单闭合曲线 γ . 一般正定向.

Theo 6.9. $g: [a, b] \rightarrow \mathbb{C}$ 连续, $|\int_a^b g(t) dt| \leq \int_a^b |g(t)| dt$.

Theo 6.10. ["Maximum Length"]. $\gamma: [a, b] \rightarrow \mathbb{C}$ 分段光滑, f 连续, $\forall z \in \gamma^*, \exists M \in \mathbb{R}$ s.t. $|f(z)| \leq M$, 则

$$|\int_{\gamma} f(z) dz| \leq M \cdot \Lambda(\gamma^*), \text{ 其中 } \Lambda(\gamma^*) = \int_a^b |\gamma'(t)| dt.$$

Theo 6.11.

$f: U \rightarrow \mathbb{C}$ 连续, $\forall$ 围道 $\gamma \subset U, \int_{\gamma} f(z) dz = 0$.

$\Rightarrow \exists F$ 在 U 解析 s.t. $\forall z \in U, F'(z) = f(z)$.

Coro.

$f: U \rightarrow \mathbb{C}$ 连续, $\forall$ 围道 $\gamma \subset U, \int_{\gamma} f(z) dz = 0$.

$\Rightarrow \forall$ 围道 $\gamma^* \subset U, \int_{\gamma^*} f(z) dz = 0$.

Chap 7.

Theo 7.1 [Cauchy Theorem].

圆道 $\gamma: [a, b] \rightarrow \mathbb{C}$, $f: U \rightarrow \mathbb{C}$ 在 U 解析, $I(\gamma) \cup \gamma^* \subset U$, 则

$$\int_{\gamma} f(z) dz = 0.$$

Theo 7.2 [Cauchy Theo for 三角圆道].

三角圆道 T , $f: U \rightarrow \mathbb{C}$ 在 U 解析, $I(T) \cup T^* \subset U$, 则

$$\int_T f(z) dz = 0.$$

Coro. [Cauchy for $\square$].

圆道 $\gamma: [a, b] \rightarrow \mathbb{C}$, $f: U \rightarrow \mathbb{C}$ 在 U 解析, $I(\gamma) \cup \gamma^* \subset U$, 则

$$\int_{\gamma} f(z) dz = 0.$$

Coro. [Cauchy for 多角].

多角圆道 $\gamma: [a, b] \rightarrow \mathbb{C}$, $f: U \rightarrow \mathbb{C}$ 在 U 解析, $I(\gamma) \cup \gamma^* \subset U$, 则

$$\int_{\gamma} f(z) dz = 0.$$

Theo 7.3.

分段光滑 $\gamma_1, \gamma_2: [a, b] \rightarrow \mathbb{C}$, $\gamma_1(a) = \gamma_2(a)$, $\gamma_1(b) = \gamma_2(b)$, $\gamma_1(t_1) \neq \gamma_2(t_2)$, $(t_1, t_2 \in (a, b))$.

$f: U \rightarrow \mathbb{C}$ 在 U 解析, $\gamma_1^*, \gamma_2^* \subset U$, 则

$$\int_{\gamma_1} f(z) dz = \int_{\gamma_2} f(z) dz.$$

Theo 7.4. [Deformation Theo].

两圆道 $\gamma_2 \subset I(\gamma_1)$, $f: U \rightarrow \mathbb{C}$ 在 U 解析, $\gamma_1, \gamma_2 \subset U$, 则

$$\int_{\gamma_1} f(z) dz = \int_{\gamma_2} f(z) dz.$$

* Deformation Theo 允许中间有奇点, 而 Cauchy Theo 必须圆道中间及圆道都解析才得到 $\int_{\gamma} f(z) dz = 0$.

所以 Deformation Theo 的 $\int_{\gamma_1} f(z) dz = \int_{\gamma_2} f(z) dz$ 有意义, 在同个无奇点区域内执行.

Chap 8.

Theo 8.1 [Cauchy Integral]

$f: U \rightarrow \mathbb{C}$ 在 U 解析. 国道 $I(\gamma) \cup \gamma^* \subset U$. $\Rightarrow \forall a \in I(\gamma)$, $f(a) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{z-a} dz$.

Theo 8.2.

$f: U \rightarrow \mathbb{C}$ 在 U 解析. 国道 $I(\gamma) \cup \gamma^* \subset U$. $\Rightarrow \forall a \in I(\gamma)$, $f'(a) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{(z-a)^2} dz$.

Coro. 解析 $f \in C^\infty$ 有任意阶导.

Theo 8.3.

$f: U \rightarrow \mathbb{C}$ 在 U 解析. 国道 $I(\gamma) \cup \gamma^* \subset U$. $\Rightarrow \forall a \in I(\gamma)$, $\forall n \geq 1$ 有 $f^{(n)}$

$$f^{(n)}(a) = \frac{n!}{2\pi i} \int_{\gamma} \frac{f(z)}{(z-a)^{n+1}} dz.$$

Theo 8.4. [Morera's Theo].

$f: U \rightarrow \mathbb{C}$ 连续. $\forall$ 国道 $\gamma \subset U$, $\int_{\gamma} f(z) dz = 0 \Rightarrow f$ 在 U 解析.

Theo 8.5. [Liouville's Theo].

全有界 f 是常数函数.

Theo 8.6. 多项 $p \in \mathbb{C}[z]$ 阶数 $n \geq 1$, $p(z) = a_n z^n + a_{n-1} z^{n-1} + \dots + a_1 z + a_0$, $a_n \neq 0$, 则 $\exists a \in \mathbb{C}$ s.t. $p(a) = 0$.

Theo 8.7. [代数基本定理]

$p(z) \in \mathbb{C}[z]$, $p(z) = a_n z^n + \dots + a_1 z + a_0$, $a_n \neq 0$, 则 $\exists \beta, \alpha_1, \dots, \alpha_n \in \mathbb{C}$ s.t.
 $p(z) = \beta(z - \alpha_1) \dots (z - \alpha_n)$.

Theo 8.8. [Indefinite Integral Theo].

f 在国道 $\gamma^* \cup I(\gamma)$ 解析 $\Rightarrow \exists$ 解析 F s.t. $\forall z \in I(\gamma)$, $F'(z) = f(z)$.

Theo 8.9.

开集 D 不包括 $\{0\} \Rightarrow \exists$ 在 D 解析的 F s.t. $e^{F(z)} = z, z \in D$.

Theo 8.10.

f 在 $N(c, R)$ 解析, 则在 $N(c, R)$ 内

$$f(z) = \sum_{n=0}^{\infty} a_n (z-c)^n, \quad a_n = \frac{f^{(n)}(c)}{n!}$$

Def. [Taylor series]. $f(z) = \sum_{n=0}^{\infty} a_n (z-c)^n$.

Theo 8.11.

解析 f 在 c 的 Taylor 为 $\sum_{n=0}^{\infty} a_n (z-c)^n$. $R > 0$. 圆道 γ^* 有 $I(\gamma) \cup \gamma^* \subset N(c, R)$, 则

$$\int_{\gamma} \left(\sum_{n=0}^{\infty} a_n (z-c)^n \right) = \sum_{n=0}^{\infty} \left(\int_{\gamma} a_n (z-c)^n \right).$$

Chap 9. 在奇点的 Taylor.

Theo 9.1 [Laurent series].

f 在 $D'(c, R)$ 解析, 则 $\exists a_n \in \mathbb{C}$, $n \in \mathbb{Z}$ s.t. $\forall z \in D'(c, R)$,

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z-c)^n.$$

$0 < r < R$, 圆环圆道 $\gamma_r^* = K(a, r)$ 有

$$a_n = \frac{1}{2\pi i} \int_{\gamma_r} \frac{f(w)}{(w-c)^{n+1}} dw.$$

Def. [主成分, principal part].

f 在 c 的主成分: $g(z) = \sum_{n=-\infty}^{-1} a_n (z-c)^n$.

[余项, residue].

f 在 c 的余项: $\text{res}(f, c) = a_{-1} \Rightarrow \int_{\gamma_r} f(z) dz = 2\pi i \cdot a_{-1} = 2\pi i \cdot \text{res}(f, c)$.

留数定理.

Def.

f 在奇点 c 有 Laurent series.

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z-c)^n.$$

①. $\forall$ 无限多 $n < 0$, $a_n \neq 0 \Rightarrow c$ 本性奇点, essential.

②. 非本性奇点 c 阶数 $\text{ord}(f, c) = n$, n 为最小的使 $a_n \neq 0$ 的数.

$\text{ord}(f, c) \geq 0 \Rightarrow c$ 可去奇点, removable.

$\text{ord}(f, c) < 0 \Rightarrow c$ 极点, pole, 阶数 $n = |\text{ord}(f, c)|$.

Prop. f, g 在 c 有限阶,

$$\text{ord}(fg, c) = \text{ord}(f, c) + \text{ord}(g, c)$$

$$\text{ord}\left(\frac{1}{f}, c\right) = -\text{ord}(f, c).$$

$$\text{ord}(f, c) < \text{ord}(g, c) \Rightarrow \text{ord}(f+g, c) = \text{ord}(f, c)$$

求 Laurent.

①. 用 $g=0$ 或展开求出表达式

(注意: 是否满足使用 $g=0$ 要求)

②. 整理成标准型

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z-c)^n.$$

求 residue.

①. 判别奇点类型, 得极点阶数 m .

$$\textcircled{2}. g(z) = (z-c)^m \cdot f(z).$$

$$\text{res}(f, c) = \frac{g^{(m-1)}(c)}{(m-1)!}$$

Theo 9.2. [Cauchy - Weierstrass].

$d \in \mathbb{C}$, f 在 C 本性奇 $\Rightarrow \forall \delta > 0, \exists \epsilon > 0, \exists z \in D'(C, \delta)$ s.t.

$$|f(z) - d| < \epsilon.$$

留数定理.

Theo 9.3. [Residue Theo]. 闭曲线 γ 及其内部奇点, residue.

f 在圆道 $L(\gamma) \cup \gamma^* \subset U$ 解析, 极点 $c_1, \dots, c_m \in L(\gamma)$, 则

$$\int_{\gamma} f(z) dz = 2\pi i \cdot \sum_{k=1}^m \text{res}(f, c_k).$$

Theo 9.4.

f 在 c -阶极点 $\Rightarrow \text{res}(f, c) = \lim_{z \rightarrow c} (z-c) f(z),$

$$f(z) = \frac{a}{z-c} + O(1) \text{ 时有 } \text{res}(f, c) = a.$$

Theo 9.5.

g, h 在 C 解析, $f(z) = \frac{g(z)}{h(z)}, h(c) = 0, g(c) \neq 0, h'(c) \neq 0$, 则

$$\text{res}(f, c) = \frac{g(c)}{h'(c)}.$$

Theo 9.6.

f 在 c - m 阶极点 $\Rightarrow f(z) = \sum_{n=-m}^{\infty} a_n (z-c)^n, a_{-m} \neq 0,$

$$\text{res}(f, c) = \frac{1}{(m-1)!} g^{(m-1)}(c), \text{ 其中 } g(z) = (z-c)^m f(z).$$

Theo 9.7.

f 在圆道 $L(\gamma) \cup \gamma^* \subset U$ 亚纯, $Q = \{q \in L(\gamma) \mid \text{ord}(f, q) \neq 0\}$ 有限, 则

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f'(z)}{f(z)} dz = \sum_{q \in Q} \text{ord}(f, q) \rightarrow \text{零点阶数} - \text{极点阶数和}.$$

1. ~~f 不解析~~

1. 定义. f 不解析 $\Rightarrow \int_{\gamma} f(z) dz = \int_a^b f(\gamma(t)) \cdot \gamma'(t) dt$.

2. Fundamental Theo. $\exists$ 解析 $F' = f \Rightarrow \int_{\gamma} f(z) dz = F(\gamma(b)) - F(\gamma(a))$.

3. Cauchy's Theo. γ 围道, f 在 $I(\gamma) \cup \gamma^*$ 解析 $\Rightarrow \int_{\gamma} f(z) dz = 0$

4. Cauchy's Integral. γ 围道, $f = \frac{g(z)}{z-c}$, g 在 $I(\gamma) \cup \gamma^*$ 解析, $c \in I(\gamma)$
 $\Rightarrow \int_{\gamma} f(z) dz = \int_{\gamma} \frac{g(z)}{z-c} dz = 2\pi i \cdot g(c)$

5. Cauchy's Integral. γ 围道, $f = \frac{g(z)}{(z-c)^n}$, g 在 $I(\gamma) \cup \gamma^*$ 解析, $c \in I(\gamma)$
 $\Rightarrow \int_{\gamma} f(z) dz = \int_{\gamma} \frac{g(z)}{(z-c)^n} dz = 2\pi i \cdot \frac{g^{(n-1)}(c)}{(n-1)!}$

[General Theo.]

6. Laurent expansion + Deformation Theo. γ 围道, f 在 $I(\gamma) \cup \gamma^*$ 亚纯且
唯一奇点 $c \in I(\gamma)$

$\Rightarrow \int_{\gamma} f(z) dz = 2\pi i \cdot \text{res}(f, c)$.

$g(z) = (z-c) f(z)$
 $\text{res}(f, c) = g'(c)$.

7. Residue Theo. γ 围道, f 在 $I(\gamma) \cup \gamma^*$ 亚纯, 极点 $c_1, \dots, c_m \in I(\gamma)$
 $\Rightarrow \int_{\gamma} f(z) dz = 2\pi i \cdot \sum_{k=1}^m \text{res}(f, c_k)$.

$g(z) = (z-c)^n f(z)$
 $\text{res}(f, c) = \frac{g^{(n-1)}(c)}{(n-1)!}$